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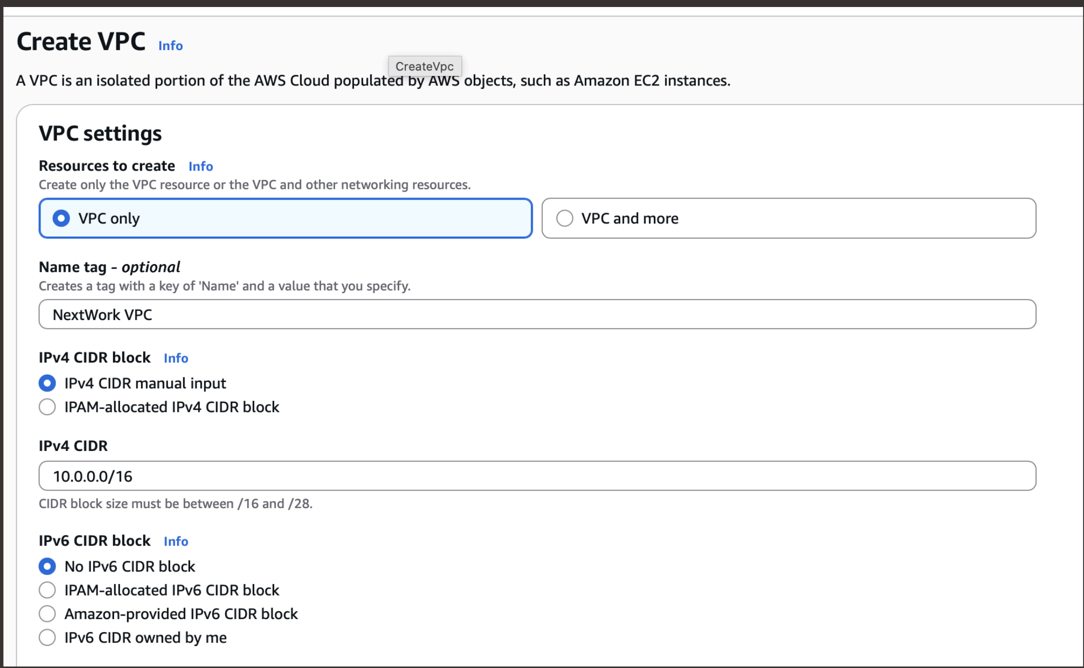
Build a Virtual Private Cloud (VPC)

Nabil Ahmed



Introducing Today's Project!

In this project, I will demonstrate how to build a Virtual Private Cloud and how to use an internet gateway to connect it to the internet. I'm doing this project to learn more about AWS services and also apply some of the networking fundemental knowledge I know into practice.

What is Amazon VPC?

Amazon VPC is a isolated network where you can manage and organise resources and it is useful because you can control which resources can be public and which resources can be made private so that there is good security.

In today's project, I used Amazon VPC to create a public subnet where resources within that subnet can be allocated a public IP address and with the use of an innternet gateway can be connected to the internet.

Personal reflection

This project took me around 1 hour as I was working with services I have not used before and also refreshing my networking knowledge especially about the difference between private and public subnets and when to use each of them.

One thing I didn't expect in this project was using IP CIDR addresses as I have only learnt about them in theory and never knew this is one way they are applied in the real world.

Virtual Private Clouds (VPCs)

What I did in this step

In this step, I will create a VPC because I need one to begin this project.

How VPCs work

VPCs are a logically isolated section of a public cloud where you can launch and manage your cloud resources. It provides control over networking, including IP addresses, subnets, routing, and security, allowing your resources to be securely isolated from other users.

Why there is a default VPC in AWS accounts

There was already a default VPC in my account ever since my AWS account was created. This is because AWS automatically creates a default VPC in each Region when a new AWS account is set up, allowing users to launch resources immediately without having to configure their own network.

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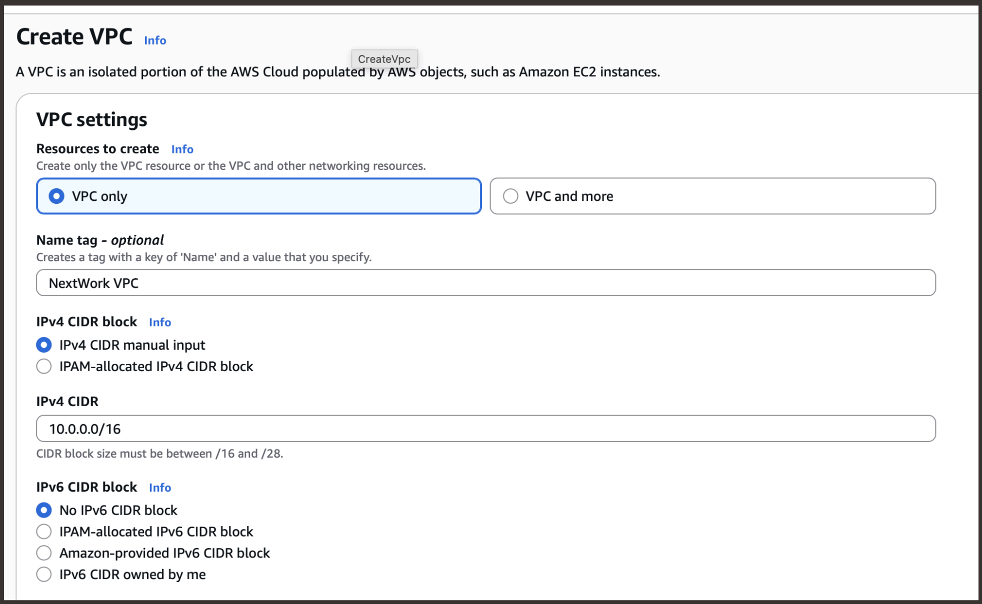


Defining IPv4 CIDR blocks

To set up my VPC, I had to define an IPv4 CIDR block, which specifies the range of IP addresses available within the VPC. The IP is split into two parts, the network portion and host portion I am using /16 which means the first 16 bits are used to identify the network and the last 16 bits are available for host addresses within that network.

Subnets

What I did in this step

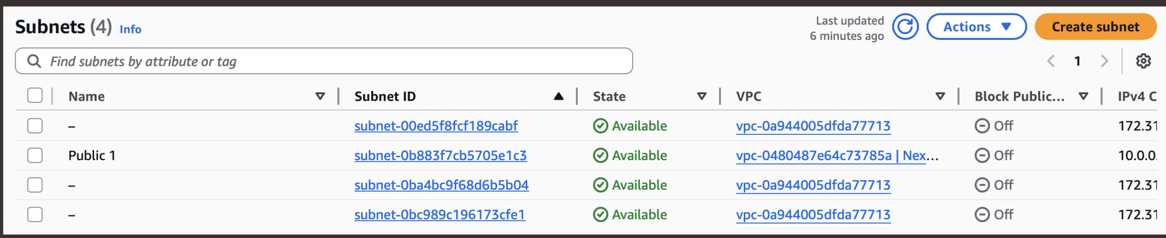
In this step, I will create a subnet for my VPC because resources must be launched into a subnet. A subnet divides the VPC into smaller networks and helps organise resources while controlling network access.

Creating and configuring subnets

Subnets are subdivisions within a VPC that allow you to organise and isolate resources with similar characteristics or purposes. My account already contained default subnets, with one subnet in each Availability Zone of the Region. Availability Zones are physically separate data centres (or groups of data centres) within an AWS Region. Distributing resources across multiple Availability Zones improves availability and fault tolerance because if one Availability Zone experiences an outage, resources in another Availability Zone can continue operating.

Public vs private subnets

The difference between public and private subnets is that resources in a private subnet cannot be accessed directly from the internet. Private subnets are typically used to host sensitive resources, such as databases and internal application servers. A public subnet, on the other hand, allows resources to communicate with the internet through an Internet Gateway. Public-facing resources, such as web servers, are usually placed in a public subnet so users can access them. For a subnet to be considered public, its route table must contain a route to an Internet Gateway.



Subnets (4) Info

Last updated 6 minutes ago | Actions | Create subnet

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 C
–	subnet-00ed5f8fcf189cabf	Available	vpc-0a944005dfda77713	Off	172.31
Public 1	subnet-0b883f7cb5705e1c3	Available	vpc-0480487e64c73785a | Nex...	Off	10.0.0.
–	subnet-0ba4bc9f68d6b5b04	Available	vpc-0a944005dfda77713	Off	172.31
–	subnet-0bc989c196173cfe1	Available	vpc-0a944005dfda77713	Off	172.31

Auto-assigning public IPv4 addresses

Once I created my subnet, I enabled auto-assign public IPv4 addresses. This setting ensures that any EC2 instance launched into the subnet is automatically assigned a public IPv4 address, so that it can be accessed from the internet.

Internet gateways

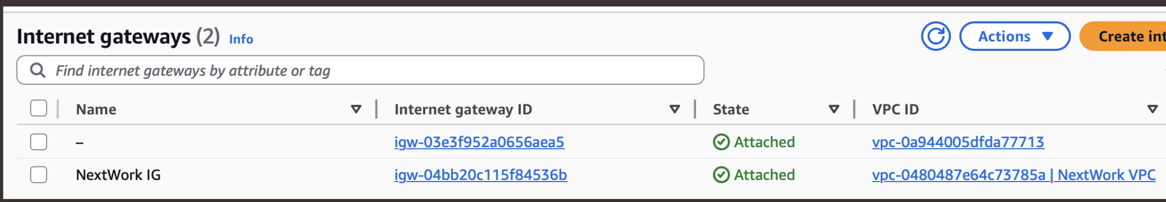
What I did in this step

In this step, I will set up an internet gateway because I need that in order to set up a route for the public subnet to be able to connect to the internet.

Setting up internet gateways

An Internet Gateway allows a public subnet to connect to the internet. It enables EC2 instances with public IP addresses to communicate with users on the internet, as long as the appropriate routing and security rules are in place.

Attaching an internet gateway to a VPC means that resources in the public subnet can now communicate with the internet. If I missed this step resources in the public subnet would not be able to communicate with the internet, even if they had public IP addresses, because there would be no route connecting the VPC to the internet.


Internet gateways (2) Info
Actions
Create int
Find internet gateways by attribute or tag
Name
Internet gateway ID
State
VPC ID
–
igw-03e3f952a0656aea5
Attached
vpc-0a944005dfda77713
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igw-04bb20c115f84536b
Attached
vpc-0480487e64c73785a | NextWork VPC

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